Paper 6 Analysis Table

Category	Details
Research Paper Title	Revolutionizing Breast Ultrasound Diagnostics with EfficientNet-B7 and Explainable AI
Objective	To develop a high-performance deep learning model using EfficientNet-B7 with explainable AI to improve breast cancer detection from ultrasound images.
Problem Statement	Breast ultrasound diagnosis is operator-dependent and prone to variability, while many AI models lack interpretability, limiting their clinical adoption. (MDPI)
Model Used	- Deep Learning: EfficientNet-B7 (Transfer Learning) - CNN-based classification architecture
Programming / Tools	➤ Python ➤ TensorFlow / Keras ➤ Image preprocessing & augmentation ➤ GPU-based training
Dataset	Breast ultrasound datasets (BUSI or similar clinical datasets)
Explainability	✔ Explainable AI techniques (e.g., Grad-CAM) used to visualize tumor regions
Pros	✔ Very high accuracy using EfficientNet-B7 ✔ Strong feature extraction capability ✔ Explainability improves trust ✔ Suitable for medical decision support
Cons	✘ High computational cost (heavy model) ✘ Requires powerful hardware (GPU) ✘ Limited real-world deployment validation
Challenges	⚠ Large model size (complex training) ⚠ Dataset dependency ⚠ Overfitting risk ⚠ Clinical integration challenges
Conclusion	EfficientNet-B7 combined with explainable AI significantly improves diagnostic accuracy and provides interpretable results, making it promising for clinical applications.

Critical Comment	This paper is strong because it uses a **state-of-the-art model (EfficientNet-B7) + XAI**, but it lacks segmentation and advanced interpretability methods. Your FYP (UltraXAI) can improve by adding **Swin-UNet segmentation + Grad-CAM++ + SHAP**, making your system more complete and advanced.